its diameter, find the equation for its inductance.

11. An inductor of inductance L carries an electric current i. How much energy is stored while establishing the current in it?

12. Show that the mutual inductance between a pair of coils is same ($M_{12} = M_{21}$).

13. How will you induce an emf by changing the area enclosed by the coil?

14. Show mathematically that the rotation of a coil in a magnetic field over one rotation induces an alternating emf of one cycle.

15. Elaborate the standard construction details of AC generator.

16. Explain the working of a single-phase AC generator with necessary diagram.

17. How are the three different emfs generated in a three-phase AC generator? Show the graphical representation of these three emfs.

18. Explain the construction and working of transformer.

19. Mention the various energy losses in a transformer.

20. Give the advantage of AC in long distance power transmission with an example.

21. Find out the phase relationship between voltage and current in a pure inductive circuit.

22. Derive an expression for phase angle between the applied voltage and current in a series RLC circuit.

23. Define inductive and capacitive reactance. Give their units.

24. Obtain an expression for average power of AC over a cycle. Discuss its special cases.

25. Show that the total energy is conserved during LC oscillations.

26. Prove that energy is conserved during electromagnetic induction.

27. Compare the electromagnetic oscillations of LC circuit with the mechanical oscillations of block-spring system to find the expression for angular frequency of LC oscillators mathematically.

IV. Numerical problems

1. A square coil of side 30 cm with 500 turns is kept in a uniform magnetic field of 0.4 T. The plane of the coil is inclined at an angle of 30° to the field. Calculate the magnetic flux through the coil. (Ans: 9 Wb)

2. A straight metal wire crosses a magnetic field of flux 4 mWb in a time 0.4 s. Find the magnitude of the emf induced in the wire. (Ans: 10 mV)

3. The magnetic flux passing through a coil perpendicular to its plane is a function of time and is given by $\Phi_B = (2t^3 + 4t^2 + 8t + 8)\ Wb$. If the resistance of the coil is 5 Ω, determine the induced current through the coil at a time t = 3 second. (Ans: 17.2 A)

4. A closely wound coil of radius 0.02 m is placed perpendicular to the magnetic field. When the magnetic field is changed from 8000 T to 2000 T in 6 s, an emf of 44 V is induced. Calculate the number of turns in the coil.

(Ans: 35 turns)

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